SOLIVA GUARD

E-Commerce Platform

PROJECT DOCUMENTATION

- **Prepared For:** Soliva Guard
- **Developed By:** TATVEIN Technologies
- **Version:** 1.0
- **Status:** Production Ready & Delivered
- **Date:** June 2026

Brand Identity & Design Guidelines

SOLIVA GUARD BRAND IDENTITY (PRIMARY BRAND)

Based on the visual design of the official **Soliva Guard** logo:

- **The Circle & Aura:** The orange circle represents the sun, outdoor life, and protection. It forms a stylized protective aura around the central emblem.
- **The ‘S’ Glyph:** The elegant off-white letter **’S’** stands for Soliva. Its clean serif typography reflects premium style, reliability, and high-performance protection.

- **Luxury & Premium Appeal:** The combination of vibrant orange and warm cream projects trust, wellness, and luxury activewear.

TATVEIN TECHNOLOGIES IDENTITY (TECHNOLOGY PARTNER)

- **Developer Sign-off:** Represented by the circuit-mandala gold logo. Used strictly as the engineering signature and technology partner badge, reflecting system architecture, automation, and backend engineering.

COLOR SPECIFICATIONS

- **Primary Color (Soliva Orange):** `#ff7a00` - Used for primary accents, links, active buttons, and UV weather highlights.
- **Secondary Color (Dark Navy):** `#0f172a` - Used for headings, table headers, and structural borders.
- **Light Gray:** `#f8fafc` - Used for page content backdrops and alternating table rows.
- **Accent Color (TATVEIN Gold):** `#b88445` - Used only in footers and partner reference elements.

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Executive Summary

Dear Soliva Guard Operations Team,

We are pleased to deliver the complete, production-ready **Soliva Guard E-Commerce & Weather-Aware Protection Platform**. This document outlines the business values, user interfaces, backend systems, and infrastructure designed and built for your high-performance protective wear brand.

THE OPPORTUNITY & THE SOLUTION

Soliva Guard stands at the intersection of luxury protective apparel and modern technology. To cater to commuters, riders, and outdoor professionals navigating extreme Indian microclimates, we have built a storefront that is not just a standard catalog, but an **intelligent assistant**. By fetching local weather indices (UV exposure, air quality, temperature), the platform automatically guides users to the ideal fabric selections—boosting customer engagement and conversion rates.

SYSTEM AT A GLANCE

The ecosystem has been delivered as a unified product consisting of a front-end storefront, a highly secure backend API server, and a comprehensive administration dashboard:

- **32 Application Screens & Interfaces:** Providing detailed pages for home, catalog, shopping cart, profile management, support ticket messaging, and order tracking.
- **72 Secure API Endpoints:** Operating behind the scenes to coordinate login sessions, database writes, payment validations, and PDF generations.
- **117 Reusable UI Components:** Optimized codebase modules ensuring rapid load times, smooth GSAP and Framer Motion layout transitions, and interactive WebGL canvas animations.
- **Turnkey Automation:** Fully automated invoicing, receipt creation, order timeline updates, and 15-minute unpaid stock recovery queues.

This handover document serves as a complete reference for your day-to-day operations, business planning, and future technical maintenance.

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1. Project Delivery Summary

The Soliva Guard platform has been successfully developed and delivered as a complete e-commerce ecosystem including customer storefront, secure backend infrastructure, payment processing, customer management, and administrative control systems.

DELIVERED STATISTICS

Item	Count
Application Screens & Interfaces	32
Reusable Components	117
API Endpoints	72
Controllers	11
Services	12
Database Models	10

2. What Has Been Delivered

The client has received a complete, ready-to-operate digital e-commerce ecosystem consisting of:

- ✓ **Complete E-commerce Website:** A premium, fully responsive online storefront showcasing products with smooth transitions and animations.
- ✓ **Customer Authentication System:** Secure customer account creation and login mechanisms.
- ✓ **Google Sign-In Integration:** One-click secure login using Google accounts.
- ✓ **OTP Verification System:** 6-digit email codes (One-Time Passwords) to verify customer emails during signup and password recovery.
- ✓ **Product Catalog System:** Dynamic list of items with categories, filter tags, search bars, and pricing.
- ✓ **Shopping Cart System:** A cart drawer that updates items and subtotal pricing as users shop.
- ✓ **Checkout System:** A step-by-step checkout portal to collect shipping and billing details.

- ✓ **Razorpay Payment Integration:** Secure online payment processing accepting UPI, credit/debit cards, and wallets.
- ✓ **Customer Profile System:** Personal dashboard managing addresses, preferences, and notifications.
- ✓ **Order Tracking System:** Visual timeline showing fulfillment progress (packed, shipped, out for delivery).
- ✓ **Admin Dashboard:** Central management console displaying revenue, customer counts, and open orders.
- ✓ **Product Management Panel:** Administrative controls to add, edit, or archive products and adjust stock counts.
- ✓ **Order Management Panel:** System to review order details, track payments, and update shipping progress.
- ✓ **Customer Management Panel:** Dashboard showing user telemetries (login locations, devices) with account block controls.
- ✓ **Analytics Dashboard:** Business performance charts showing sales volume and growth trends.
- ✓ **Email Notification System:** Automatic transactional emails for signup OTPs, welcomes, and purchase invoices.
- ✓ **Security Layer:** Rate limiters, encryption, and data sanitizers protecting the business from online threats.
- ✓ **Outdoor Intelligence Recommendation Engine:** Location-aware dashboard using local weather, UV, and air quality indices to suggest appropriate products.

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3. Pages Delivered

The platform contains **32 application screens and interfaces** and **5 layouts/wrappers** in the frontend client.

CLIENT-FACING PAGE INVENTORY

Page Name	Route	Purpose	User Type
Home Page	`/`	Storefront home showcasing interactive 3D particle animations, fabric values, and featured products.	Public (Guest / Customer)
Collection Page	`/collection`	Storefront catalog displaying categories with grid systems, filters, sorting, and search.	Public (Guest / Customer)
Product Details	`/products/$slug`	Detailed product overview displaying images, comparison pricing, descriptions, and ratings.	Public (Guest / Customer)
Shopping Cart	`/cart`	Overview of items in cart, quantities, subtotals, and checkout navigation.	Public (Guest / Customer)
Checkout Portal	`/checkout`	Stepped form handling shipping inputs, address verification, and payment triggers.	Protected (Customer)
Payment Success	`/payment-success`	Redirect page indicating payment has been captured.	Protected (Customer)
Order Confirmed	`/order-confirmed`	Confirmation page showing invoice download links and receipts.	Protected (Customer)

Page Name	Route	Purpose	User Type
Customer Profile	`/profile`	User dashboard managing account profiles, preferences, and notifications.	Protected (Customer)
Order History	`/profile` *(Orders tab)*	Table of past orders, statuses, and navigation.	Protected (Customer)
Order Details	`/profile/orders/$id`	Detailed history showing item summaries, transaction info, and invoice downloads.	Protected (Customer)
Order Tracking	`/track/$orderId`	Live order timeline detailing tracking events (e.g. packed, shipped, out for delivery).	Protected (Customer)
Contact Desk	`/contact`	Message submission form to communicate with customer service.	Public (Guest / Customer)
Brand Story	`/story`	Editorial view showcasing the company's background, vision, and fabric values.	Public (Guest / Customer)
Technology / Specs	`/technology`	Interactive explanation of fabric mechanics, air filtration layers, and UV testing results.	Public (Guest / Customer)
Support Helpdesk	`/support`	Customer ticket center displaying ticket threads and replies.	Protected (Customer)
Terms & Conditions	`/terms`	View detailing service agreements and delivery terms.	Public (Guest / Customer)

Page Name	Route	Purpose	User Type
Login Page	`/login`	Customer login view (supports traditional credential checks and Google authentication).	Public (Guest / Customer)
Register Page	`/register`	User sign-up view with OTP verification.	Public (Guest / Customer)
Forgot Password	`/forgot-password`	Form initiating account recovery by sending a 6-digit OTP.	Public (Guest / Customer)
Reset Password	`/reset-password`	Form to change account password after OTP validation.	Public (Guest / Customer)
Admin Login	`/soliva-admin`	Restricted administration access screen.	Public (Admin)
Admin Dashboard	`/admin/dashboard`	Main metrics desk showing gross sales, order graphs, and queue stats.	Protected (Admin)
Admin Products Grid	`/admin/products`	Table listing all catalog items with actions to edit or delete.	Protected (Admin)
Admin Create Product	`/admin/products/new`	Product creation panel with image upload controls.	Protected (Admin)
Admin Edit Product	`/admin/products/$productId/edit`	Catalog update screen.	Protected (Admin)
Admin Orders Table	`/admin/orders`	Table listing client orders, status filters, and dispatch status updates.	Protected (Admin)

Page Name	Route	Purpose	User Type
Admin Order Detail	`/admin/orders/$orderId`	Detailed order summary displaying invoices, payment IDs, and timeline logs.	Protected (Admin)
Admin Customer List	`/admin/customers`	Audit list showing registered users, geolocation telemetries, and account locking switches.	Protected (Admin)
Admin Customer Detail	`/admin/customers/$customerId`	Detailed user logs showing session metrics, active devices, and location maps.	Protected (Admin)
Admin Inventory Grid	`/admin/inventory`	Live inventory table with fast-edit stock counters.	Protected (Admin)
Admin Payments Log	`/admin/payments`	Overview of transaction receipts, payments, and invoice codes.	Protected (Admin)
Admin Analytics Desk	`/admin/analytics`	Sales statistics, registration growth, and product category charts.	Protected (Admin)
Admin Settings	`/admin/settings`	Portal to manage company variables and server statuses.	Protected (Admin)

REUSABLE LAYOUT ELEMENTS

- **Root Wrapper:** `__root.tsx` - Governs site navigation, footers, search overlays, and common styles.
- **Auth Layout:** `_auth/route.tsx` - Directs access checks to keep authenticated pages secure.
- **Public Layout:** `_public/route.tsx` - Serves as the outer container for customer-facing navigation elements.

- **Admin Layout:** `admin/route.tsx` - Verifies administrative privileges before rendering dashboard screens.
- **Admin Products Layout:** `admin/products.tsx` - Handles nested routes for product administration.

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4. API Infrastructure

BUSINESS SUMMARY

An **API (Application Programming Interface)** acts as a digital courier that allows different parts of a software application to speak to each other. The Soliva Guard platform contains **72 API endpoints** that link user interface actions (like clicking “pay” or typing a registration request) to database operations securely behind the scenes.

- **Public routes** are open to any browser connection (e.g., retrieving products).
- **Protected routes** require the customer to be logged in with a secure access token.
- **Admin routes** require admin credentials to prevent unauthorized data access.

API ENDPOINTS LISTING

Method	Endpoint	Purpose	Access Level
POST	`/api/v1/auth/register`	Register new account and send verification OTP.	Public
POST	`/api/v1/auth/verify-email`	Validate registration OTP and create user document.	Public
POST	`/api/v1/auth/resend-otp`	Generate and resend registration/recovery OTP.	Public (Rate Limited)
POST	`/api/v1/auth/login`	Authenticate credentials and issue session cookie.	Public (Rate Limited)
POST	`/api/v1/auth/google`	Validate Firebase Google Sign-In token and login.	Public
POST	`/api/v1/auth/admin-login`	Authenticate admin credentials and issue session.	Public (Rate Limited)
POST	`/api/v1/auth/refresh`	Renew expired JWT session cookie.	Public
POST	`/api/v1/auth/forgot-password`	Request password reset OTP.	Public (Rate Limited)
POST	`/api/v1/auth/verify-forgot-otp`	Verify password recovery OTP.	Public
POST	`/api/v1/auth/reset-password`	Set new password using verified OTP.	Public
GET	`/api/v1/auth/logout`	Revoke current session.	Protected

Method	Endpoint	Purpose	Access Level
GET	`/api/v1/auth/logout-all`	Revoke all active sessions for the user.	Protected
GET	`/api/v1/auth/me`	Fetch authenticated user profile.	Protected
POST	`/api/v1/auth/location`	Update user location and telemetry.	Protected
GET	`/api/v1/auth/sessions`	List active sessions, devices, and locations.	Protected
DELETE	`/api/v1/auth/sessions/:sessionId`	Revoke a specific session by ID.	Protected
DELETE	`/api/v1/auth/sessions`	Revoke all other active sessions.	Protected
POST	`/api/v1/users/me/avatar`	Upload and set profile avatar.	Protected
DELETE	`/api/v1/users/me/avatar`	Remove profile avatar.	Protected
GET	`/api/v1/products/`	Query products catalog with filters and search.	Public
GET	`/api/v1/products/:idOrSlug`	Fetch single product detail by ID or Slug.	Public
POST	`/api/v1/products/admin/new`	Create a new product.	Admin
PUT	`/api/v1/products/admin/:id`	Update product details.	Admin
DELETE	`/api/v1/products/admin/:id`	Delete product.	Admin
GET	`/api/v1/categories/`	Fetch all product categories.	Public
GET	`/api/v1/profile/`	Fetch customer profile, preferences, and	Protected

Method	Endpoint	Purpose	Access Level
		wishlist.	
PUT	`/api/v1/profile/lifestyle`	Update user lifestyle options.	Protected
PUT	`/api/v1/profile/preferences`	Update product color and usage preferences.	Protected
PUT	`/api/v1/profile/notifications`	Toggle email, WhatsApp, and push flags.	Protected
POST	`/api/v1/profile/wishlist`	Add product to customer wishlist.	Protected
DELETE	`/api/v1/profile/wishlist/:productId`	Remove product from wishlist.	Protected
GET	`/api/v1/profile/recent`	Fetch recently viewed products.	Protected
POST	`/api/v1/profile/recent`	Log product view to history.	Protected
GET	`/api/v1/profile/addresses`	List customer delivery addresses.	Protected
POST	`/api/v1/profile/addresses`	Add new address to address book.	Protected
PUT	`/api/v1/profile/addresses/:id/default`	Set address as default.	Protected
PUT	`/api/v1/profile/addresses/:id`	Update address details.	Protected
DELETE	`/api/v1/profile/addresses/:id`	Delete address from address book.	Protected
GET	`/api/v1/profile/tickets`	List support tickets.	Protected
GET	`/api/v1/profile/tickets/:id`	Fetch ticket details and message thread.	Protected

Method	Endpoint	Purpose	Access Level
POST	`/api/v1/profile/tickets`	Submit support ticket.	Protected
POST	`/api/v1/profile/tickets/:id/reply`	Send message reply in ticket thread.	Protected
GET	`/api/v1/cart/`	Fetch shopping cart items.	Protected
POST	`/api/v1/cart/`	Add product to cart.	Protected
DELETE	`/api/v1/cart/`	Empty shopping cart.	Protected
POST	`/api/v1/cart/merge`	Merge guest cart items into database cart.	Protected
PUT	`/api/v1/cart/:productId`	Update item quantity in cart.	Protected
DELETE	`/api/v1/cart/:productId`	Remove item from cart.	Protected
POST	`/api/v1/orders/`	Create a new order and Razorpay intent.	Protected
POST	`/api/v1/orders/verify-payment`	Verify Razorpay payment signature.	Protected
GET	`/api/v1/orders/me`	Fetch past orders for the customer.	Protected
GET	`/api/v1/orders/:id`	Fetch order details.	Protected
PUT	`/api/v1/orders/:id/cancel`	Cancel order.	Protected
GET	`/api/v1/orders/admin/orders`	List all orders.	Admin
PUT	`/api/v1/orders/admin/orders/:id`	Update order delivery status.	Admin
GET	`/api/v1/admin/dashboard`	Fetch dashboard analytics.	Admin
GET	`/api/v1/admin/products`	List all products.	Admin

Method	Endpoint	Purpose	Access Level
GET	`/api/v1/admin/customers/stats`	Fetch customer registration stats.	Admin
GET	`/api/v1/admin/customers/export`	Export customer records.	Admin
GET	`/api/v1/admin/customers`	List all users with login telemetries.	Admin
GET	`/api/v1/admin/customers/:id`	Fetch customer profile details.	Admin
PATCH	`/api/v1/admin/customers/:id/status`	Block or unblock a customer.	Admin
GET	`/api/v1/admin/inventory`	List all inventory items.	Admin
PATCH	`/api/v1/admin/inventory/:productId`	Update inventory stock count.	Admin
GET	`/api/v1/admin/payments`	List transaction logs.	Admin
GET	`/api/v1/admin/analytics`	Fetch sales charts and metrics.	Admin
POST	`/api/v1/payment/webhook`	Process Razorpay webhook messages.	Public
POST	`/api/v1/payment/test-order`	Temporary test route for order verification.	Public
GET	`/api/v1/system/health`	Check MongoDB, Redis, and queue health.	Public
POST	`/api/v1/upload`	Upload product images.	Admin
POST	`/api/v1/contact/`	Submit general contact messages.	Public
POST	`/api/v1/contact/feedback`	Submit feedback with up to 5 image	Public

Method	Endpoint	Purpose	Access Level
		attachments.	

5. Database Structure

BUSINESS SUMMARY

The database acts as the digital filing cabinet of your business. **Models** define the structure of the data folders (e.g., Users, Products, Orders) stored inside the database, ensuring information is saved in a uniform format. High-speed indexes act as quick-lookup tabs to search customer and purchase records in milliseconds.

TECHNICAL DATABASE SCHEMA

Model	Database Collection	Business Role	Fields & Relationships
User	`users`	Stores accounts, security controls, location telemetries, and sessions.	• `name`, `username`, `email` (Unique) • `password` (Hashed, select: false) • `googleId` (OAuth link) • `role` (user/admin) • `status` (active/blocked) • `lastLoginIP`, `lastLoginCity`, `lastLoginState` • `sessions` (Subdocument array: IP hash, device, browser, expiry)
Product	`products`	Stores products, prices, stock counts, and reviews.	• `name`, `slug` (Unique) • `description`, `price`, `compareAtPrice` • `currency` (e.g. INR) • `images` (URLs) • `category`, `tags`, `featured` • `stock` (Available count) • `ratings`, `reviewsCount` • `user` (Ref: `User` creator)
Order	`orders`	Records order details, transactions, tracking histories, and PDF files.	• `shippingInfo` (Full address snapshot) • `orderItems` (Array of items: name, price, qty, product ref) • `user` (Ref: `User` customer) • `paymentInfo` (Razorpay ID, signature, status) • `itemsPrice`, `taxPrice`, `shippingPrice`, `totalPrice` • `orderStatus` (Timeline

Model	Database Collection	Business Role	Fields & Relationships
			enum) • `events` (Timeline history logs) • `invoiceNumber`, `receiptNumber`, `invoiceUrl`, `receiptUrl`
UserProfile	`userprofiles`	Stores customer preferences and lifestyle settings.	• `user` (Ref: `User` owner, Unique) • `lifestyle` (Student, commute, travel, outdoor) • `preferences` (Preferred colors, utility mode) • `wishlist` (Array of product refs, restock settings) • `notifications` (Email, WhatsApp, Push flags)
Address	`addresses`	Stores customer address entries.	• `user` (Ref: `User` owner) • `fullName`, `phone`, `houseNo`, `street`, `city`, `state`, `pincode`, `country` • `addressType` (Home, Work, Other) • `isDefault` (Boolean flag)
Cart	`carts`	Syncs active shopping carts across devices.	• `user` (Ref: `User` owner) • `cartItems` (Array: product ref, quantity) • `totalPrice` (Cart subtotal)
AuthAuditLog	`authauditlogs`	Logs security events and authentication attempts.	• `userId` (Ref: `User` actor) • `email` (Login input) • `action` (login success, login fail, OTP sent, etc.) • `status` (success/failure) • `ipHash`, `device`, `browser` details

Model	Database Collection	Business Role	Fields & Relationships
PendingRegistration	`pendingregistrations`	Stores unverified registration details during email OTP verification.	• `name`, `username`, `email` (Unique) • `passwordHash`, `emailOTP` • `otpExpire` (TTL index auto-delete) • `retryCount`, `lastResendAt`
RecentlyViewed	`recentlyvieweds`	Stores recently browsed products.	• `user` (Ref: `User` viewer) • `product` (Ref: `Product` browsed) • `viewCount`, `lastViewed` • Compound index: `{ user: 1, product: 1 }` (Unique)
SupportTicket	`supporttickets`	Stores support tickets and customer messages.	• `user` (Ref: `User` creator) • `order` (Ref: `Order` reference) • `subject`, `status` • `messages` (Array: sender ref, message body, sentAt)

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6. Outdoor Intelligence Dashboard

BUSINESS SUMMARY

To make product listings more relevant to buyers, the storefront includes an environmental tracker widget. The browser reads user coordinates, reverse-geocodes them into city names, and requests weather statistics. Based on these conditions, it suggests specific protective apparel (e.g. lightweight filters for high smog days or thermal fabrics for cold wind conditions).

TECHNICAL DETAILS

- **OpenWeather Integration:** Fetches local temperature, humidity, and weather conditions.
- **OpenWeather UV Index Integration:** Queries UV levels to determine local exposure values.
- **WAQI API Integration:** Fetches localized Air Quality Index (AQI) values.

- **Nominatim Geocoding:** Translates latitude/longitude coordinates into city and state strings.
- **API Key Absence Fallbacks:** If API keys are missing or requests time out, the system catches exceptions and falls back to:
  - *Weather default:* `{ temp: 30, humidity: 60, condition: 'Clear', icon: '01d', uvIndex: 5 }`
  - *AQI default:* `{ aqi: 42, status: 'Good', color: '#009966' }`
- **Recommendations Engine:** Uses UV, AQI, and Temperature indices to calculate a unified protection score (from Low to Extreme) and recommends clothing fabrics:
  - *High UV (>= 6):* “High UV Today: Full facial & neck coverage advised.”
  - *Warm Temp (> 30°C):* “Warm Weather: Opt for Cool-Tech silk blend Sunwrap.”
  - *Poor AQI (> 100):* “Poor Air Quality: Use Sunwrap with integrated filtration layer.”

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7. Order Management & Transaction Pipeline

BUSINESS SUMMARY

The system processes purchases reliably. When a customer initiates payment, the database runs all operations—verifying product stock, reserving inventory quantities, creating the invoice record, and resetting the customer cart—as a single unit of work. If the payment fails or a product runs out of stock, the transaction is rolled back, preventing incomplete orders or duplicate charges.

TECHNICAL DETAILS

The order creation process uses a transactional database pipeline:

1. **Idempotency Validation:** Validates `x-idempotency-key` in incoming request headers. If the key exists, it aborts the transaction and returns the existing order payload to prevent duplicate checkouts.
2. **Inventory Check & Decrement:** Queries product items. If stock is available, it atomically decrements the quantity (`$inc: { stock: -quantity }`). If stock is insufficient, the transaction rolls back.
3. **Order Generation:** Generates the database order record in `pending` status, linked to a Razorpay order ID.
4. **Cart Clear:** Empties the user’s active shopping cart.
5. **Commit/Rollback:** Commits the transaction if all steps succeed. On any validation error, it aborts the transaction and rolls back the inventory.

6. **Fulfillment State Machine:** Enforces valid status transitions:
    - `pending` -> `paid` or `cancelled`
    - `paid` -> `confirmed`, `packed`, `cancelled`, or `refunded`
    - `confirmed` -> `packed`, `shipped`, `cancelled`, or `refunded`
    - `packed` -> `shipped`, `cancelled`, or `refunded`
    - `shipped` -> `delivered`, `cancelled`, or `refunded`
    - `delivered` -> `refunded`

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8. Authentication, Privacy & System Security

BUSINESS SUMMARY

The platform protects user data using modern encryption standards. Login tokens are locked inside secure browser cookies that cannot be read by external scripts, and access is monitored to prevent brute-force login attempts or bot activity.

TERMINOLOGY DEFINITIONS

- **JWT (JSON Web Token):** A secure, encrypted digital identity card used to confirm login status.
- **HttpOnly Cookie:** A browser cookie that blocks client-side script access, protecting session keys from theft.
- **Bcrypt Hashing:** A cryptographic security function used to hash passwords, rendering them unreadable in the database.
- **Rate Limiting:** Server rule setting a limit on the number of requests an IP address can make in a given timeframe to prevent server overload.

TECHNICAL DETAILS

- **JWT & Cookie Authentication:** Access credentials are encrypted in JWT format and stored inside HTTP-Only, Secure, SameSite cookies. This prevents client-side scripts from reading the token.
- **Session Management & Device Tracking:** Tracks active sessions in the database using IP hashes, device footprints, browser details, and timestamps, allowing users to audit active devices and revoke unauthorized access.
- **Google OAuth Single Sign-In:** Google OAuth is initiated on the client using the Firebase SDK. The backend verifies the token via the Firebase Admin SDK, creating a passwordless user profile if one doesn’t exist.

- **Secure OTP Validation:** Generates a 6-digit cryptographically secure OTP, storing a SHA256 hash in a temporary table. This hash must match the user's input before registration or password recovery is complete.
- **Brute-Force Account Locking:** Locks user accounts for 15 minutes after 5 consecutive failed login attempts, writing the block parameters directly to the user document.
- **Strict Rate Limiting middleware:** Uses `express-rate-limit` to restrict API traffic. Login routes are limited to 20 requests per 15 minutes, and OTP requests are limited to 5 requests per 15 minutes.
- **NoSQL Injection Cleaner Middleware:** Middleware parses all inputs and strips any parameters prefixed with `$` to protect MongoDB query parameters.
- **XSS Cleaners & HPP Prevention:** Sanitization routines filter HTML tags from text inputs. The backend also blocks duplicate HTTP parameters (HPP) to prevent route hijack attempts.

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9. Payment System Integration

BUSINESS SUMMARY

The payment system integrates directly with Razorpay, a certified payment gateway. Customer payments are verified using security signatures to prevent fraud. If a checkout is abandoned, the reserved stock is automatically returned to the store catalog after 15 minutes.

TERMINOLOGY DEFINITIONS

- **HMAC (Hash-based Message Authentication Code):** A digital seal used to verify the integrity and authenticity of payment callback signatures.
- **Webhook Listener:** A server script that listens for real-time transaction updates sent by Razorpay (e.g. payment confirmed).

TECHNICAL DETAILS

- **Razorpay SDK Integration:** Integrates the Razorpay SDK to create payment orders. Supports fallback configurations, generating mock orders in local development environments.
- **Secure Payment Verification:** Verifies the Razorpay payment signature by generating a local HMAC SHA256 signature using the order and payment IDs and comparing it to the signature header.

- **Reconciling Webhooks:** Processes incoming payment events (`order.paid`, `payment.captured`). It parses signatures, matches payment records to internal orders, and updates order states to `paid` in the background if a user closes the browser during checkout.
- **Order Stock Allocation:** Uses MongoDB transactions to handle checkouts. It checks product stock, atomically updates inventory, creates the order, and empties the cart as a single unit of work to prevent duplicate allocations.
- **Refund Management:** The order status machine allows state updates to `refunded` and automatically restores inventory stock. However, the backend does **not** call Razorpay's API to initiate a financial refund transaction. Financial refunds must be processed manually in the Razorpay dashboard.
- **Delayed Stock Recovery Queue:** If an order remains unpaid, a delayed job queue running via BullMQ cancels the pending order and restores the reserved product stock after 15 minutes.

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10. Email & Automated Notifications

BUSINESS SUMMARY

The system sends automatic transactional emails (verification codes, welcome greetings, support receipts, invoices) without slowing down the storefront. Tasks are enqueued and processed by background workers.

TECHNICAL DETAILS

- **OTP Email Flow:** Generates a 6-digit OTP during signup or password recovery, enqueuing a background email job (`enqueueEmail('registrationOTP')` or `enqueueEmail('forgotPasswordOTP')`) to deliver the validation code.
- **Welcome Email Flow:** Upon email verification or first Google login, enqueues the `welcomeEmail` background task containing user preference tips.
- **Contact & Support Requests:** The `/api/v1/contact` endpoint submits contact messages, sending a copy to the support desk (reply-to customer email) and an automated receipt copy to the customer. The support feedback form supports up to 5 attachments, sending them as inline attachments and cleaning up temporary disk storage once sent.
- **Order Notifications:** Success verification triggers background PDF tax invoice and receipt generation, then enqueues the `orderConfirmation` email containing absolute links to download the files.

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11. Product Catalog

The platform database comes preloaded with **5 flagship protective wraps** priced in Indian Rupees (INR) with high-quality mock product image paths.

1. **Soliva AirShield Wrap** (slug: `soliva-airshield-wrap`, tags: `['Advanced UV Defense', 'Breathable', 'Full Coverage']`, stock: `100`, price: `799 INR`, compare-at price: `1200 INR`)
2. **Soliva Urban Veil** (slug: `soliva-urban-veil`, tags: `['Advanced UV Defense', 'Lightweight', 'Daily Wear']`, stock: `100`, price: `799 INR`, compare-at price: `1200 INR`)
3. **Soliva HeatGuard** (slug: `soliva-heatguard`, tags: `['Heat Reflective', 'Advanced UV Defense', 'All-Day Comfort']`, stock: `100`, price: `799 INR`, compare-at price: `1200 INR`)
4. **Soliva MotionCover** (slug: `soliva-motioncover`, tags: `['Advanced UV Defense', 'Stretch-Soft', 'Indian Climate']`, stock: `100`, price: `799 INR`, compare-at price: `1200 INR`)
5. **Soliva AirLite Shield** (slug: `soliva-airlite-shield`, tags: `['Ultra-Light', 'Advanced UV Defense', 'Breathable']`, stock: `100`, price: `799 INR`, compare-at price: `1200 INR`)

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12. Business Benefits

- **Improved Customer Experience:** Offers a fast, modern storefront that automatically recommends products based on local weather conditions.
- **Faster Order Management:** Features interactive timelines and automatic invoice/receipt PDF generation.
- **Secure Online Payments:** Integrates Razorpay with signature verification to prevent transaction tampering.
- **Automated Email Communication:** Asynchronous email queues send notifications without slowing down the server.
- **Inventory Control:** Automatically releases stock from abandoned carts after 15 minutes, preventing stock lockups.
- **Administrative Oversight:** Admin console to monitor transactions, block suspicious users, and track revenue.
- **Business Analytics:** Interactive charts display sales and customer growth trends.
- **Scalable Architecture:** Built on Node.js and Redis, capable of handling high traffic and concurrent sessions.

13. Client Handover Notes



The delivered platform includes all core systems required for day-to-day business operations.

Administrative access allows management of:

- **Products:** Add or archive products, upload images, and set pricing.
- **Orders:** Update package delivery progress and trigger cancellations.
- **Customers:** Review customer login histories and block suspicious accounts.
- **Payments:** Reconcile transaction logs with Razorpay orders.
- **Analytics:** Track sales performance, monthly signups, and popular items.
- **Inventory:** Directly adjust stock quantities to maintain availability.

The system is configured for future scalability and maintenance.

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14. Maintenance & Server Hosting Charges

Hosting and maintenance are billed **annually**, as per the Hosting & Website Maintenance Quotation (#TAT-2026-06-28-001) and the Server Configuration & Deployment Invoice (#TAT-2026-06-29-002):

Description	Plan	Amount
Annual Cloud Server Hosting — 4 vCPU (Intel®/AMD), 8 GB RAM, 80 GB NVMe SSD	1 Year	**₹12,000/year** (≈ ₹1,000/month, billed as a single annual payment)
Website Maintenance (Annual Plan)	1 Year	**₹9,600/year**
Complimentary Technical Support	30 Days	Included
Daily Backup & Server Monitoring	—	Included
Ubuntu Server Provisioning, Docker Environment, Nginx Reverse Proxy, SSL/DNS & Production Deployment	1 Year	Included
Total Annual Cost		**₹21,600/year**

The annual maintenance plan includes the following services:

- **Server Monitoring:** Regular checks to ensure frontend and backend Systemd services are running.
- **Bug Fixes:** Troubleshooting and resolving runtime application errors.
- **Security Updates:** Monitoring firewall configurations and running weekly OS system package updates.
- **Dependency Updates:** Scanning and updating npm packages to patch vulnerabilities.
- **Database Maintenance:** Regular MongoDB backups, index optimization, and storage cleanups.
- **Email Monitoring:** Checking BullMQ logs to verify email delivery.
- **Payment Monitoring:** Auditing webhooks and transaction logs.
- **Performance Monitoring:** Keeping track of Redis RAM and host storage usage.
- **Technical Support:** Consulting on server migrations or configuration adjustments.

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15. Project Investment & Value Delivered

The implementation of the Soliva Guard E-Commerce Platform provides significant strategic and operational value. TATVEIN Technologies has delivered a fully automated, secure, and customer-centric digital ecosystem designed to accelerate sales growth and streamline business administration.

STRATEGIC CAPITAL & CAPABILITIES DELIVERED:

- **Complete E-Commerce Platform:** A ready-to-operate digital store that enables direct-to-consumer (D2C) sales, eliminating reliance on third-party marketplaces.
- **32 Application Screens & Interfaces:** A comprehensive set of visual screens optimized for quick load times, smooth scrolling, and dynamic micro-interactions.
- **72 Backend API Endpoints:** A secure backend communication network that supports site operations, handling account registrations, catalog updates, and order placements.
- **Secure Authentication & Authorization:** Multi-device session controls and automatic account locking that secure customer data and build brand trust.
- **Razorpay Payment Gateway Integration:** A secure checkout flow supporting UPI, cards, and wallets with signature verification to prevent payment fraud.
- **Customer Account Management:** A dashboard allowing customers to save delivery details, track order histories, and set lifestyle and color preferences.
- **Admin Operations Dashboard:** A unified management console that provides real-time insights into gross sales, pending orders, and system health.
- **Product Management System:** An administrative tool to manage catalogs, set compare-at prices, generate SEO slugs, and upload product images.
- **Order Management System:** A transaction-safe fulfillment workflow that automates invoice/receipt PDF generation, shipment tracking, and stock recovery.
- **Email Notification Infrastructure:** An email dispatcher with automatic retries to send welcome messages, OTP codes, and transaction receipts.
- **Outdoor Intelligence Recommendation Engine:** A location-aware recommendation widget that uses weather and air quality data to suggest relevant products to users.
- **Production Deployment Configuration:** A containerized deployment setup running on a self-hosted Linux server, secured behind Cloudflare SSL and Nginx reverse proxies.

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16. Final Delivery Summary



Module	Status	Delivery Reference
Frontend Website	**Completed**	Vite Config and Root Route
Backend API	**Completed**	API Initialization App and Server Entry
Admin Panel	**Completed**	Admin Router Layout and Admin Controller Actions
Database Schemas	**Completed**	Mongoose Schema Definitions
Authentication	**Completed**	Authentication Controller and Firebase Library
Payment Integration	**Completed**	Payment Controllers and Payment Service
Email System	**Completed**	Nodemailer Utility and Email Background Worker
Security Layer	**Completed**	Centralized App Middleware and Audit Log Model
Deployment Setup	**Completed**	Systemd Service Templates and Docker Compose Configuration

Conclusion

The Soliva Guard e-commerce platform has been successfully delivered and deployed by the technology partner. The platform runs on a self-hosted Linux server, with loopback-secured database instances and structured process logging. The administrative features provide complete control over products, inventory, customers, and order dispatch workflows. We believe the platform is ready to support your business growth.

Should you wish to opt-in to the annual maintenance program or require help with server variables, please contact our support desk.

Sincerely,
TATVEIN Technologies Delivery & Integration Team